<u>CV Projects</u>

Project Scope

- Students must select only healthcare and medical imaging datasets.
- Datasets must be collected from **Data in Brief** or **Scientific Data** journals.
- Students must develop their own mini annotation tool.
- Students will perform image classification, object detection, and segmentation.
- Students must prepare a research paper as final submission.

Project Timeline

Week	Due Dates	Tasks	**Deliverables**
Week 1	12-05-2026	Dataset selection, cleaning, custom annotator development	Dataset proposal, annotator code, 20 sample annotations
Week 2	19-05-2026	Medical image annotation and classification	Annotated dataset, classification code, metrics
Week 3	26-05-2026	Object detection training and testing	Detection model, prediction outputs, mAP results “**Submit Online**”
Week 4	02-06-2026	Segmentation, evaluation, paper writing, video recording	Segmentation code, research paper, video

			demo, final package
Week 5	09-06-2026	Main Project evaluation, video recording	Segmentation code, video demo, final package

1. **Patient Monitoring System Using Cameras**

Core Idea

a) Automated system that monitors patients in hospitals using vision models.

Additional Project Ideas

- **Fall Detection System (ICU/Elderly Care)**
  - Detect when a patient falls from bed or chair using pose estimation.
- **Sleep Posture Monitoring**
  - Classify sleeping positions (supine, prone, sideways) for pressure ulcer prevention.
- **Pain Expression Detection**
  - Use facial expression analysis to estimate pain level in non-verbal patients.
- **Patient Movement Tracking**
  - Track whether a patient is mobile or stationary for long periods (bed-ridden risk analysis).
- **Tube/IV Line Monitoring**
  - Detect accidental removal or displacement of medical equipment (advanced detection task).

2. Class / Room Behavior or Anomaly Detection System

a) Core Idea

Detect abnormal or suspicious behavior in classrooms or hospital wards.

Additional Project Ideas

- **Student Attention Monitoring**
  - Detect: sleeping, distracted, mobile phone usage.
- **Cheating Detection System**
  - Identify unusual head movements or gaze direction in exams.
- **Aggressive Behavior Detection**
  - Detect fighting, pushing, or sudden physical movement spikes.
- **Crowd Density Monitoring**
  - Detect overcrowding in rooms (useful for hospital waiting areas).
- **Unauthorized Entry Detection**
  - Alert when unknown persons enter restricted zones.

3. Multi-Camera Medical Image / Surveillance Pipeline

a) Core Idea

Combine multiple cameras to improve detection accuracy and coverage.

- **Multi-Angle Patient Tracking**
  - Track a patient across multiple hospital room cameras.
- **Cross-Camera Identity Re-Identification (Re-ID)**
  - Recognize same patient across different camera views.
- **3D Pose Estimation from Multiple Cameras**
  - Reconstruct patient posture in 3D space for better diagnosis.
- **Camera Fusion for Better Detection**
  - Combine predictions from multiple cameras to reduce false alarms.

- **Blind Spot Elimination System**
  - Detect areas not covered by any camera in hospital rooms.

4. **End-to-End Intelligent Healthcare Surveillance System**

a) Core Idea

A complete AI-powered hospital monitoring ecosystem.

Additional Project Ideas

- **Real-Time ICU Monitoring Dashboard**
  - Live feed + alerts for abnormal patient activity.
- **Automated Emergency Alert System**
  - Trigger alert for fall, seizure-like movement, or distress.
- **Doctor/Nurse Activity Tracking System**
  - Monitor staff movement efficiency and patient interaction time.
- **Smart Triage System**
  - Detect severity of patient condition using visual cues (restlessness, motion level).
- **AI-Based Incident Report Generator**
  - Automatically create event logs (e.g., "Patient fall detected at 3:42 PM").

Hybrid Medical AI System

- Combine:
  - Detection (YOLO/RT-DETR)
  - Segmentation (U-Net / SAM)
  - Classification (ResNet / ViT)
- Output:
  - "Patient Risk Score Dashboard"

Behavioral Health Monitoring

- Detect:
  - Anxiety signs
  - Depression indicators (low movement, face expression)
  - Agitation levels in psychiatric wards

Predictive Alert System

- Not just detection → but prediction:
  - “Patient likely to fall in next 10 seconds”
  - “High agitation detected trend”

Evaluation Rubric

Component	Weight
Dataset Selection	5%
Annotator Development	10%
Annotation Quality	5%
Classification	10%
Detection	10%
Segmentation	10%
Research Paper	45%
Video Demo	5%